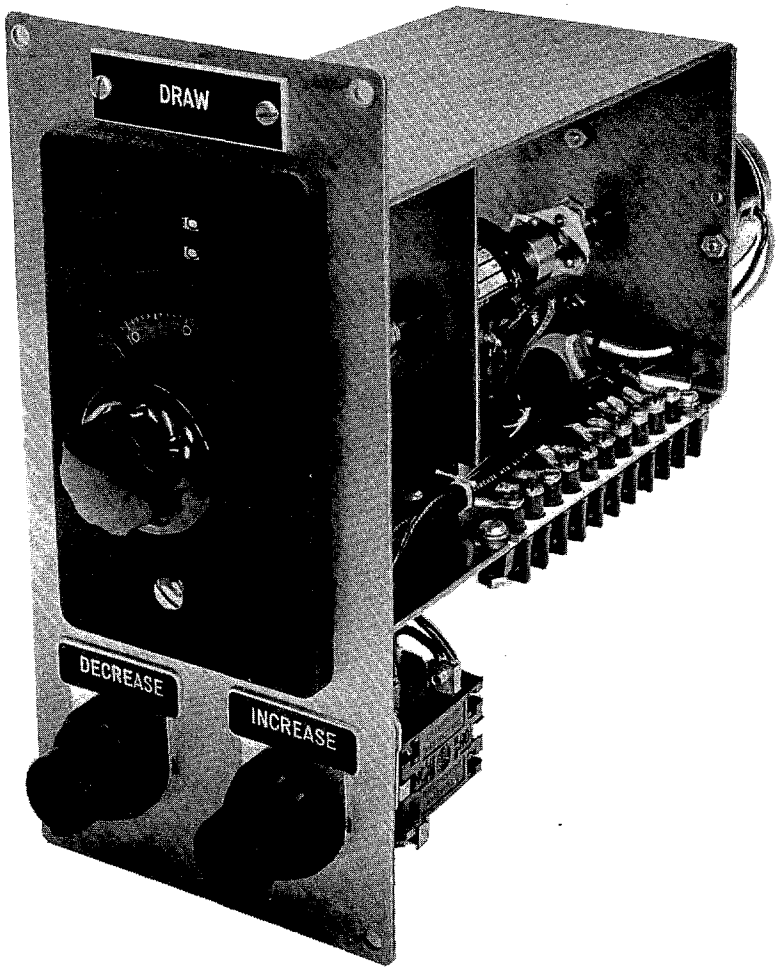
INSTRUCTIONS GEK- 7818

PROGRESSIVE DRAW MECHANISM

FOR PAPER MILL DRIVES

TYPE IC 4806



ICF-25917

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INDUSTRY CONTROL PRODUCTS DEPARTMENT

GENERAL ELECTRIC

SALEM, VA. 24153

INTRODUCTION

The Progressive Draw Mechanism is a precision, adjustable potentiometer that can have its resistance varied by either direct manual control or by a motor drive. It is normally used in motor excitation control circuits on a paper machine sectional drive to control motor speed and thus increase or decrease the amount of slack (or draw) in the paper between different paper machine sections.

DESCRIPTION

A front view photograph of the mechanism is shown on the cover of these instructions. A side view photograph is shown in Figure No. 1.

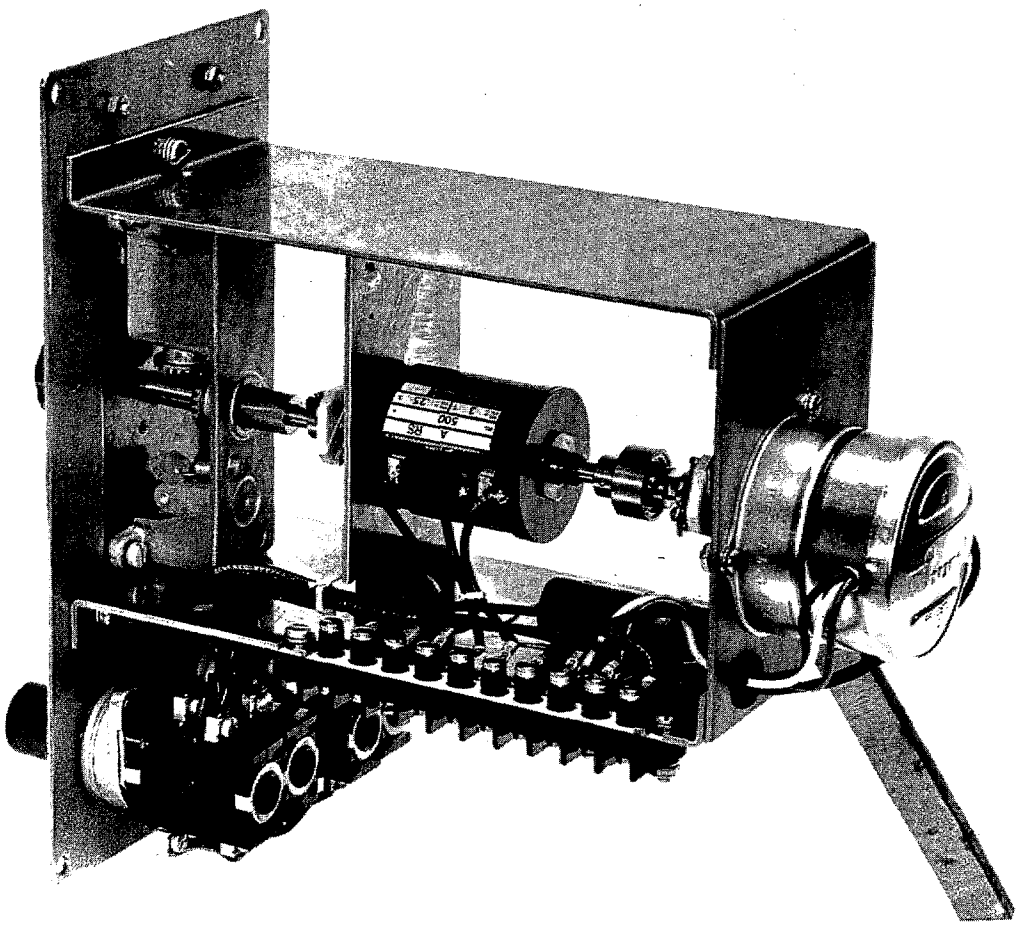


Figure No. 1 - Side View of Mechanism (ICF-25918)

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

Description (cont.)

The mechanism consists of a Helipot adjustable potentiometer with a double shaft extension. One of the shaft extensions is connected through a coupling and an electric clutch to a synchronous, reversible, 2RPM, 115 V.A.C. motor. The other shaft extension is connected through a coupling to a manually operated handwheel dial and precision indicator. Two pushbuttons are furnished for control of the motor drive. An alternate form of the mechanism is available that omits the pushbuttons and uses a separately mounted control switch or pushbuttons.

OPERATION

The Helipot can be varied in resistance by energizing the electric motor drive or by turning the manual handwheel knob. The electric clutch mechanically isolates the motor drive from the Helipot except during the time when the motor is running. An elementary and wiring diagram showing the electrical circuit for the mechanism is shown on Figures No. 2 and 3, (G.E. Drawings 148B9809 and 148B9810). pages 3 and 4.

INSTALLATION

The Progressive Draw Mechanism is normally mounted at the General Electric Company factory on an operators control desk or station. If the mechanism is to be mounted on existing equipment or on a control desk being furnished by others, a suitable location must be found and the necessary panel drilling provided. The mechanism will operate properly in any position but normally should be mounted with the face plate vertical or horizontal with the dial up to facilitate operation and observation.

Outline dimensions and mounting details are shown in Figures No. 3 and 4, (G.E. Drawings 149B2622 and 149B2623), pages 5 and 6.

The mechanism requires power at 115 volts, single phase, 60 cycles (50 cycle motors can be furnished if specified). Power consumption is approximately 5 watts.

ADJUSTMENTS - TUNE UP

The mechanism has been mechanically and electrically adjusted at the factory and should require only a visual inspection and a mechanical and electrical operation check.

Figure No. 2

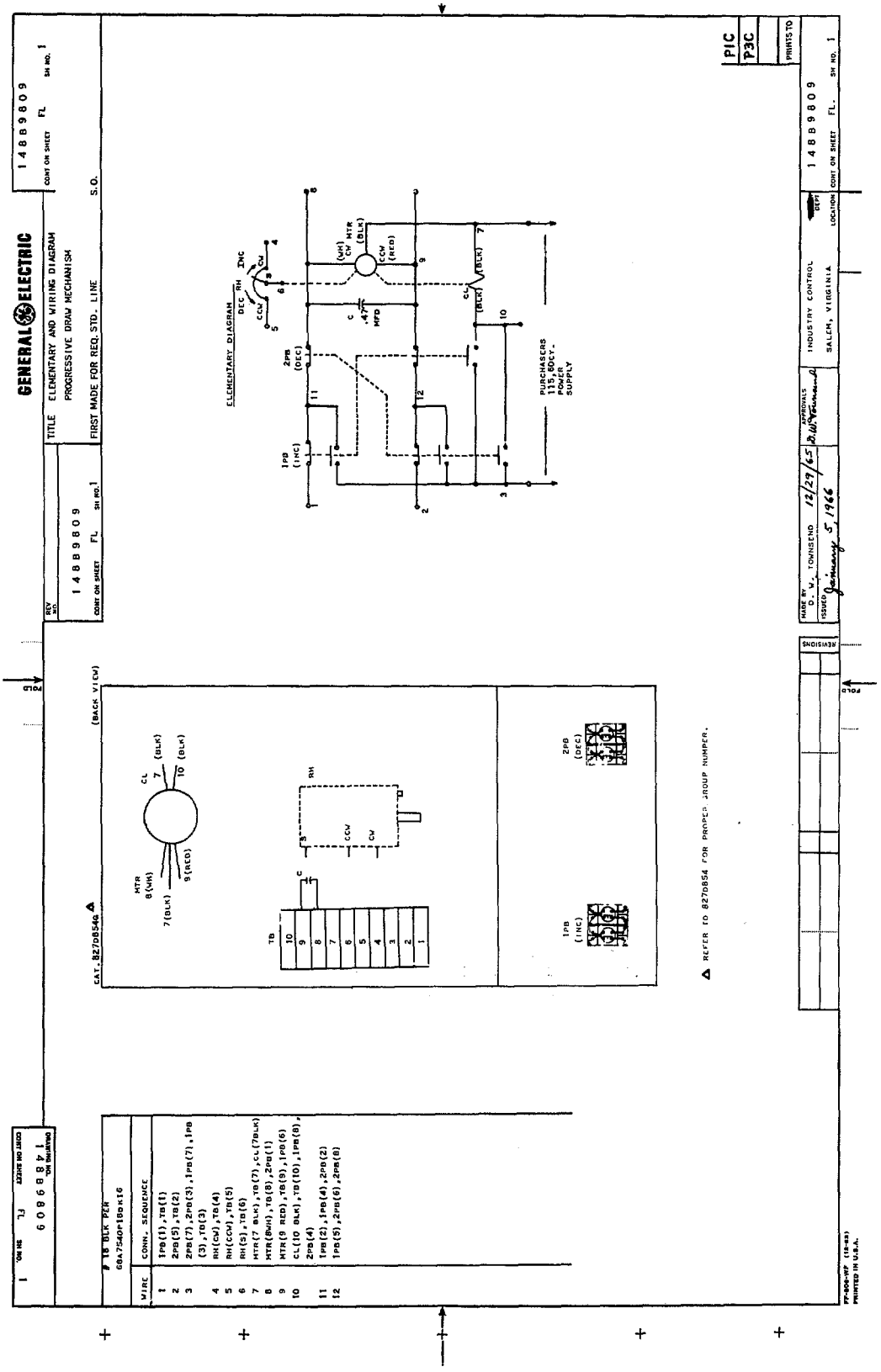

GENERAL ELECTRIC
TITLE ELEMENTARY AND WIRING DIAGRAM
PROGRESSIVE DRAW MECHANISM
FIRST MADE FOR REQ. STD. LINE
S.O.
14889809
CONT ON SHEET FL SH NO. 1
ELEMENTARY DIAGRAM
DEC RH INC
CCW CW
2PB (DEC)
1PB (INC)
C .47 MFD
(WH) CW MTR (BLK)
CCW (RED)
CL (BLK)
PURCHASERS 115 60CY. POWER SUPPLY
(BACK VIEW)
CAT. 8270B540
MTR 8(WH) 7(BLK) 9(RED)
CL 7 (BLK) 10 (BLK)
TB 10 9 8 7 6 5 4 3 2 1
CCW CW RH
2PB (DEC)
1PB (INC)
REFER TO 8270B54 FOR PROPER GROUP NUMBER.
WIRE NO. 60675401B0116
WIRE SEQUENCE
1 1PB(1),TB(1)
2 2PB(5),TB(2)
3 2PB(7),2PB(3),1PB(7),1PB(3),TB(3)
4 RH(CW),TB(4)
5 RH(CCW),TB(5)
6 RH(5),TB(6)
7 MTR(7 BLK),TB(7),CL(7BLK)
8 MTR(8WH),TB(8),2PB(1)
9 MTR(9 RED),TB(9),1PB(6)
10 CL(10 BLK),TB(10),1PB(8),2PB(4)
11 1PB(2),1PB(4),2PB(2)
12 1PB(5),2PB(6),2PB(8)
MADE BY D. W. TOWNSEND 12/29/65
ISSUED January 5, 1966
INDUSTRY CONTROL
SALEM, VIRGINIA
PIC
P3C
PRINTS TO
REVISIONS

Figure No. 3

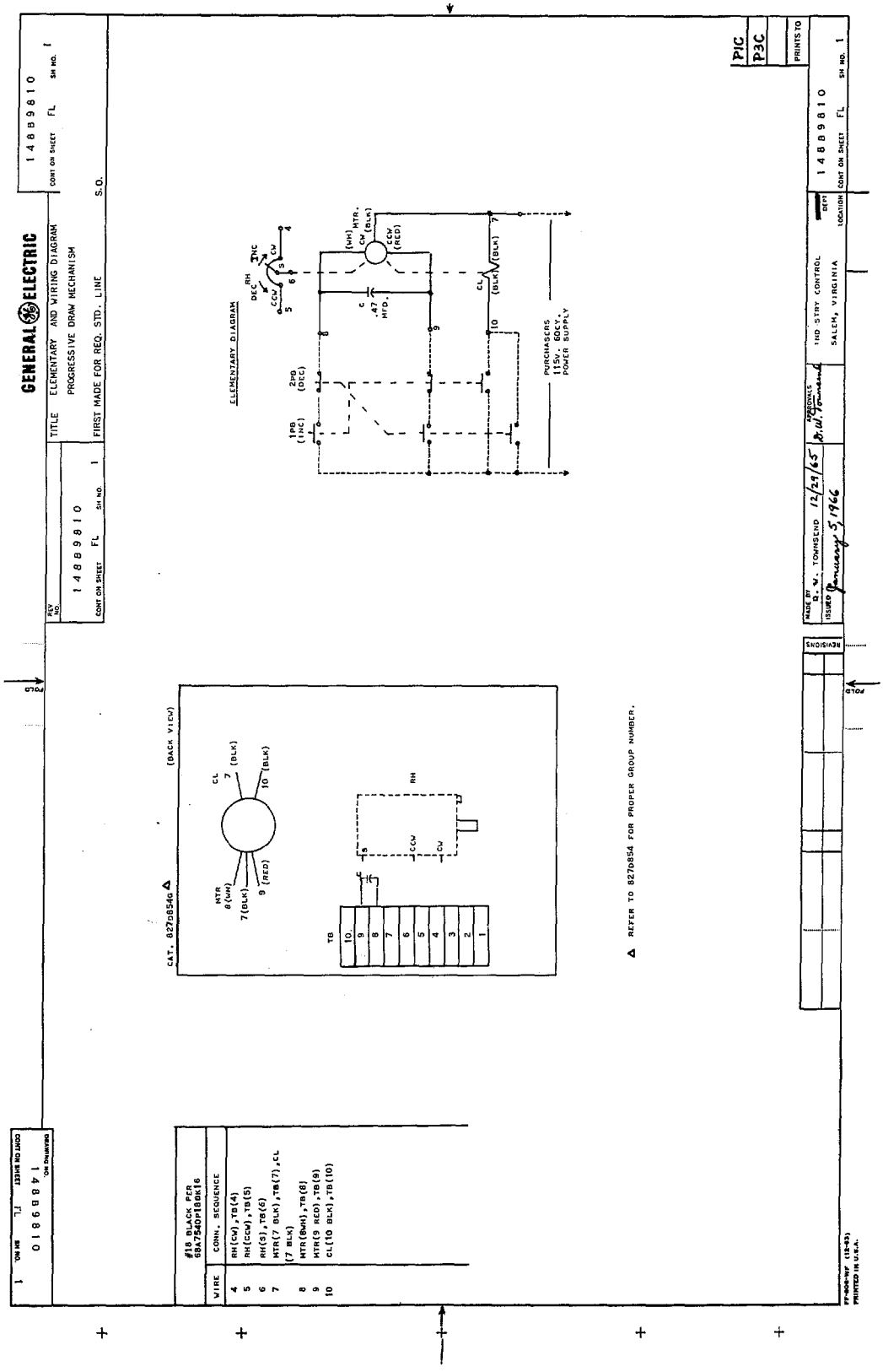

GENERAL ELECTRIC
14889810
TITLE ELEMENTARY AND WIRING DIAGRAM
PROGRESSIVE DRAW MECHANISM
FIRST MADE FOR REQ. STD. LINE
S.O.
ELEMENTARY DIAGRAM
1PB (INC)
2PB (DEC)
DEC RH
INC
CCW
CW
C .47 MFD.
MTR. (WH) CW (BLK) CCW (RED)
CL (BLK)
PURCHASERS 115v. 60CY. POWER SUPPLY
(BACK VIEW)
CAT. 827D8546 △
MTR 8(WH) 7(BLK) 9 (RED)
CL 7 (BLK) 10 (BLK)
TB
RH
CCW
CW
△ REFER TO 827D854 FOR PROPER GROUP NUMBER.
#18 BLACK PER 68A7540P18Bk16
WIRE
CONN. SEQUENCE
4 RH(CW),TB(4)
5 RH(CCW),TB(5)
6 RH(S),TB(6)
7 MTR(7 BLK),TB(7),CL (7 BLK)
8 MTR(8WH),TB(8)
9 MTR(9 RED),TB(9)
10 CL(10 BLK),TB(10)
MADE BY D. W. TOWNSEND 12/29/65
ISSUED January 5, 1966
IND STRY CONTROL
SALEM, VIRGINIA
PRINTS TO
P1C
P3C
REVISIONS

Figure No. 4

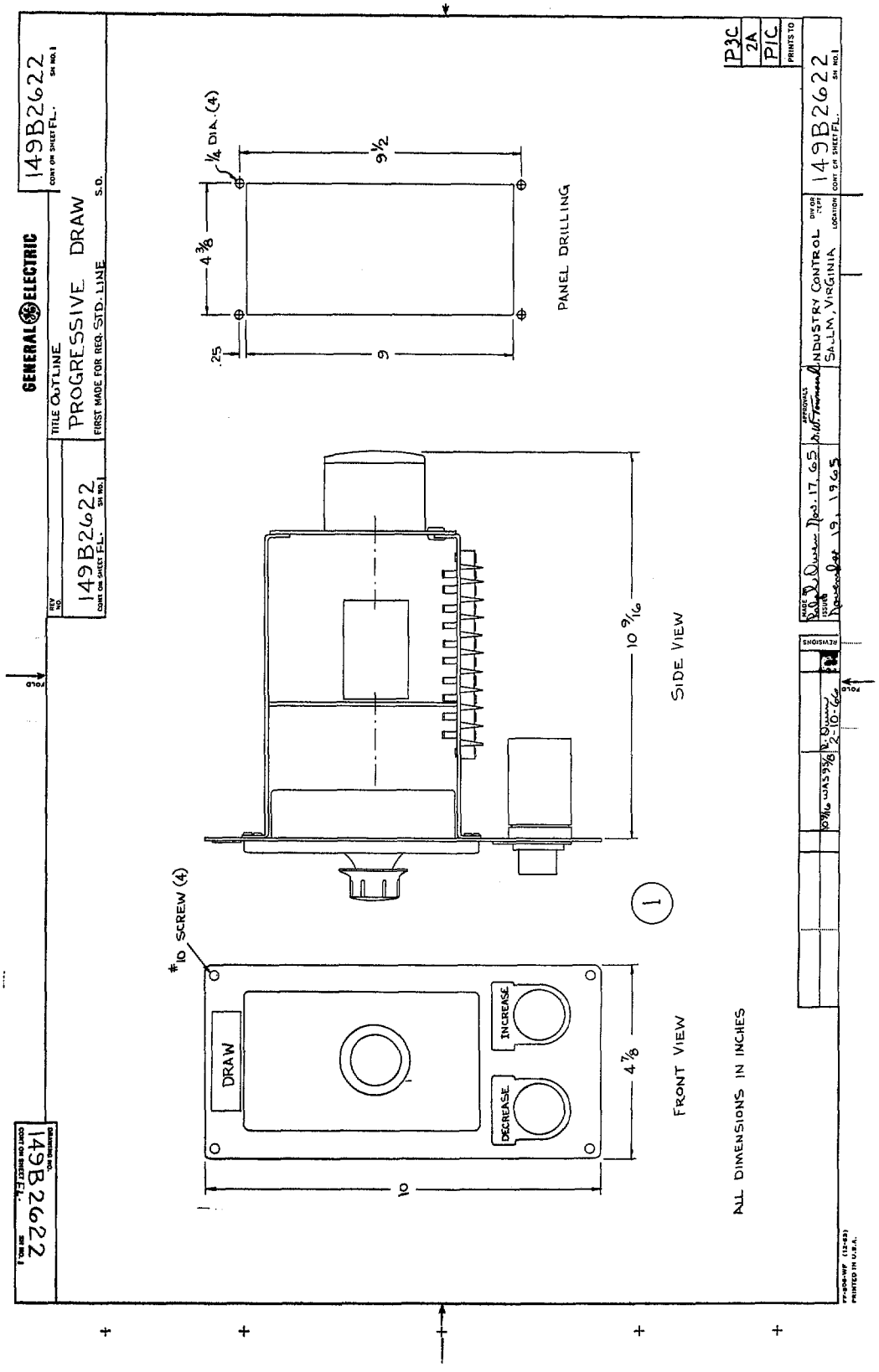
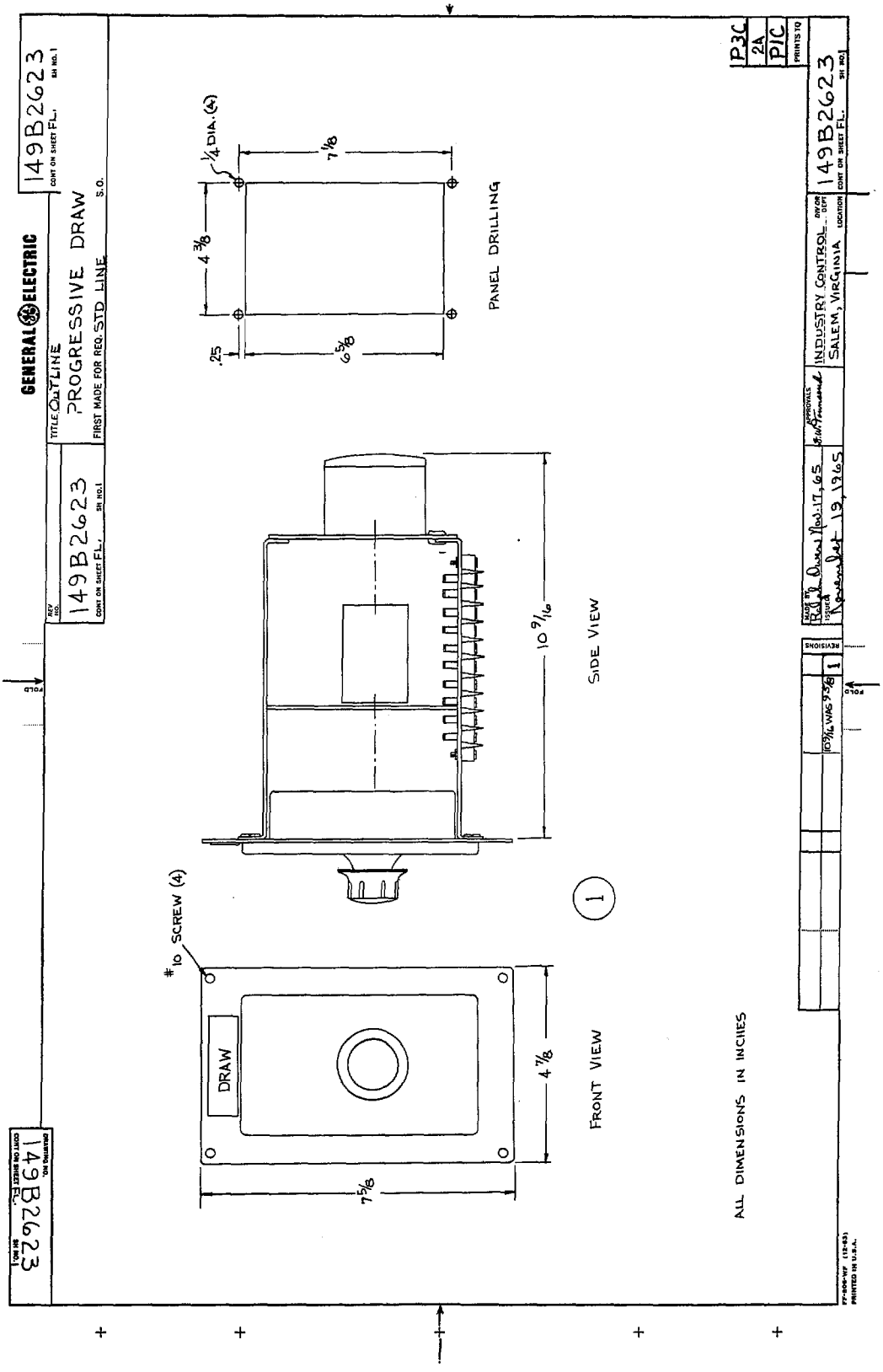

GENERAL ELECTRIC
149B2622
TITLE OUTLINE
PROGRESSIVE DRAW
FIRST MADE FOR REQ. STD. LINE
S.O.
149B2622
149B2622
PANEL DRILLING
¼ DIA. (4)
9½
4⅜
.25
9
SIDE VIEW
10 9/16
#10 SCREW (4)
DRAW
INCREASE
DECREASE
4⅞
10
FRONT VIEW
ALL DIMENSIONS IN INCHES
INDUSTRY CONTROL
SALEM, VIRGINIA
149B2622
P3C
2A
P1C
PRINTS TO

Figure No. 5


GENERAL ELECTRIC
149B2623
TITLE OUTLINE
PROGRESSIVE DRAW
FIRST MADE FOR REQ. STD LINE
S.O.
INDUSTRY CONTROL
SALEM, VIRGINIA
PANEL DRILLING
¼ DIA. (4)
7⅛
4⅜
.25
6⅝
SIDE VIEW
10 9/16
#10 SCREW (4)
DRAW
FRONT VIEW
4⅞
7⅝
ALL DIMENSIONS IN INCHES
P3C
2A
P1C
PRINTS TO

TROUBLE SHOOTING AND MAINTENANCE

The Progressive Draw Mechanism should be relatively free from trouble or maintenance problems. The mechanism should be kept clean and free from foreign matter that could interfere with its mechanical operation.

The mechanism has been carefully lined up at the factory to assure smooth operation. All shaft coupling set screws, etc. have been sealed with "Lock-Tite". This fact should be noted if it is ever necessary to disassemble the mechanism to replace a burned out or wornout Helipot.

SPARE PARTS

Spare or replacement parts for the mechanism should be ordered from the nearest General Electric Company Sales Office. Prices will be quoted on request.